NHS Prescription Cost Analysis - England 2025/26

Healthcare Data Analytics Portfolio Project

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Tools: Google Sheets | Google BigQuery | SQL
Data Source: NHS Business Services Authority (NHSBSA) - Prescription Cost Analysis
Financial Year: 2025/26

1. Executive Summary

This project analyses NHS prescription expenditure in England during 2025/26 to identify key cost drivers, prescribing patterns and areas where expenditure monitoring may provide the greatest operational value.

Using Google Sheets and Google BigQuery, the analysis evaluated approximately 1.30 billion prescription items representing £11.64 billion in expenditure, with an average cost of £8.97 per item. The Endocrine System recorded the highest expenditure among the 20 BNF chapters analysed, while Tirzepatide was identified as the highest-cost chemical substance at approximately £582.09million.

The analysis demonstrates how prescription data can be converted into structured KPIs and actionable insights to support expenditure monitoring, resource planning and evidence-based healthcare decision-making.

2. Business Problem

NHS prescribing represents substantial expenditure distributed across multiple therapeutic categories, products and chemical substances. High overall expenditure alone does not explain what is driving the cost; expenditure may result from high prescription volumes, higher costs per item, expensive individual substances, or a combination of these factors.

For decision-makers, the challenge is therefore not simply to measure total expenditure, but to determine where expenditure is concentrated, which categories are driving it, and which indicators should be monitored to identify emerging cost pressures.

Business Question

How can NHS prescription data be analysed to identify major expenditure drivers, prescribing patterns and areas that may benefit from closer cost monitoring?

3. Project Objectives

The project was designed to:

- Measure overall NHS prescription expenditure and prescription volume.
- Calculate average expenditure per prescription item.
- Compare expenditure across BNF therapeutic chapters.
- Identify the BNF chapters associated with the highest total expenditure.
- Compare cost per item across therapeutic categories.
- Identify high-cost chemical substances.
- Assess how overall expenditure is distributed across therapeutic categories.
- Develop KPIs suitable for ongoing expenditure monitoring.
- Build a dashboard that communicates the principal findings clearly to non-technical stakeholders.
- Use SQL and Google BigQuery to perform structured and reproducible analysis.

4. Dataset and Datasource

The analysis uses publicly available Prescription Cost Analysis (PCA) data published by the NHS Business Services Authority (NHSBSA) for England for the 2025/26 financial year.

The source data provides prescription expenditure and volume information organised through the British National Formulary (BNF) hierarchy.

Data Structure

BNF Chapters → BNF Sections → BNF Paragraphs → Chemical Substances

The project primarily analysed BNF chapter-level expenditure while also using chemical-substance-level information to identify individual substances associated with particularly high expenditure.

Key Variables

Variable	Purpose
Financial Year	Identifies the reporting period
BNF Chapter Code	Identifies the therapeutic chapter
BNF Chapter Name	Provides the therapeutic category
Total Items	Measures prescription volume
Total Cost	Measures prescription expenditure
Cost per Item	Enables comparison between expenditure and volume
Chemical Substance	Supports substance-level expenditure analysis

The analysis covered **20 BNF chapters** and incorporated chemical-substance-level data for more detailed investigation of expenditure concentration.

5. Data Preparation and Cleaning

The source data was prepared and structured in Google Sheets before analysis.

5.1 Data Organisation

Relevant data was separated into dedicated worksheets covering:

- National data
- BNF Chapters
- BNF Sections
- BNF Paragraphs

- Chemical Substances
- Analysis
- Presentations
- Dashboard

This created a structured analytical workbook while preserving the different levels of the BNF hierarchy.

5.2 Data Quality Review

The data was reviewed to ensure that:

- Prescription volume fields were stored in appropriate numerical formats.
- Prescription expenditure was represented consistently as numerical cost data.
- BNF categories could be correctly associated with their respective values.
- Relevant columns were clearly labelled for analysis.
- Calculated measures were suitable for KPI reporting and comparison.

5.3 Analytical Preparation

The chapter-level dataset was prepared using the following core fields:

Financial Year | BNF Chapter Code | BNF Chapter Name | Total Items | Total Cost | Cost per Item

A clean version of this dataset was subsequently imported into Google BigQuery with explicitly defined field names and data types.

This enabled the spreadsheet analysis to be complemented by structured SQL queries and provided a reproducible method for validating and extending the analysis.

6. Google Sheets Analysis

Google Sheets was used for data preparation, exploratory analysis, KPI development and dashboard creation.

The analysis focused on three interconnected dimensions:

Prescription Volume → Prescription Expenditure → Cost per Item

Considering these measures together provides greater analytical value than examining total expenditure independently, as high expenditure can be influenced by either high utilisation, high unit costs, or both.

6.1 Key Performance Indicators (KPI)

KPI	Result
Total Prescription Cost	£11,641,873,550.60
Total Prescription Items	1,297,516,707
Average Cost per Item	£8.97
Total BNF Chapters analyzed	20
Highest Cost BNF Chapter	Endocrine System
Highest Cost Chemical Substance	Tirzepatide
Highest Chemical Substance Cost	£582,093,677.53

These KPIs provide an executive-level overview of prescribing scale and expenditure before more detailed therapeutic-category analysis.

7. SQL Analysis Using BigQuery

Google BigQuery was used to extend the spreadsheet analysis and demonstrate how the same healthcare dataset could be analysed within a structured database environment.

The cleaned BNF chapter-level dataset was imported into BigQuery and queried using SQL.

7.1 Overall Prescription KPIs

SQL aggregation was used to calculate:

- Total prescription items
- Total prescription expenditure
- Average expenditure per prescription item

The SQL analysis produced approximately **1.30 billion prescription items**, **£11.64 billion in expenditure**, and an average cost of **£8.97 per item**.

7.2 BNF Chapter Expenditure Ranking

BNF chapters were ranked according to total prescription expenditure.

This enabled the analysis to distinguish therapeutic categories representing the greatest absolute expenditure and provided a basis for prioritizing areas for further investigation.

7.3 Expenditure Share Analysis

A SQL window function was used to calculate each BNF chapter's percentage contribution to overall expenditure.

This provides additional decision-making context because absolute expenditure alone does not show how strongly an individual category contributes to the total prescribing cost base.

7.4 Cost-per-Item Analysis

BNF chapters were also ranked according to average cost per prescription item.

This enabled high-volume categories to be distinguished from categories where expenditure may be influenced more strongly by comparatively expensive items.

7.5 Prescription Volume Analysis

The final SQL analysis ranked BNF chapters according to total prescription items.

Comparing prescription volume with expenditure provides a more balanced interpretation of cost drivers and reduces the risk of assuming that the categories with the greatest expenditure are necessarily the categories with the greatest utilisation.

8. Dashboard Development

A dashboard was developed in Google Sheets to translate the analytical outputs into a concise stakeholder-facing view.

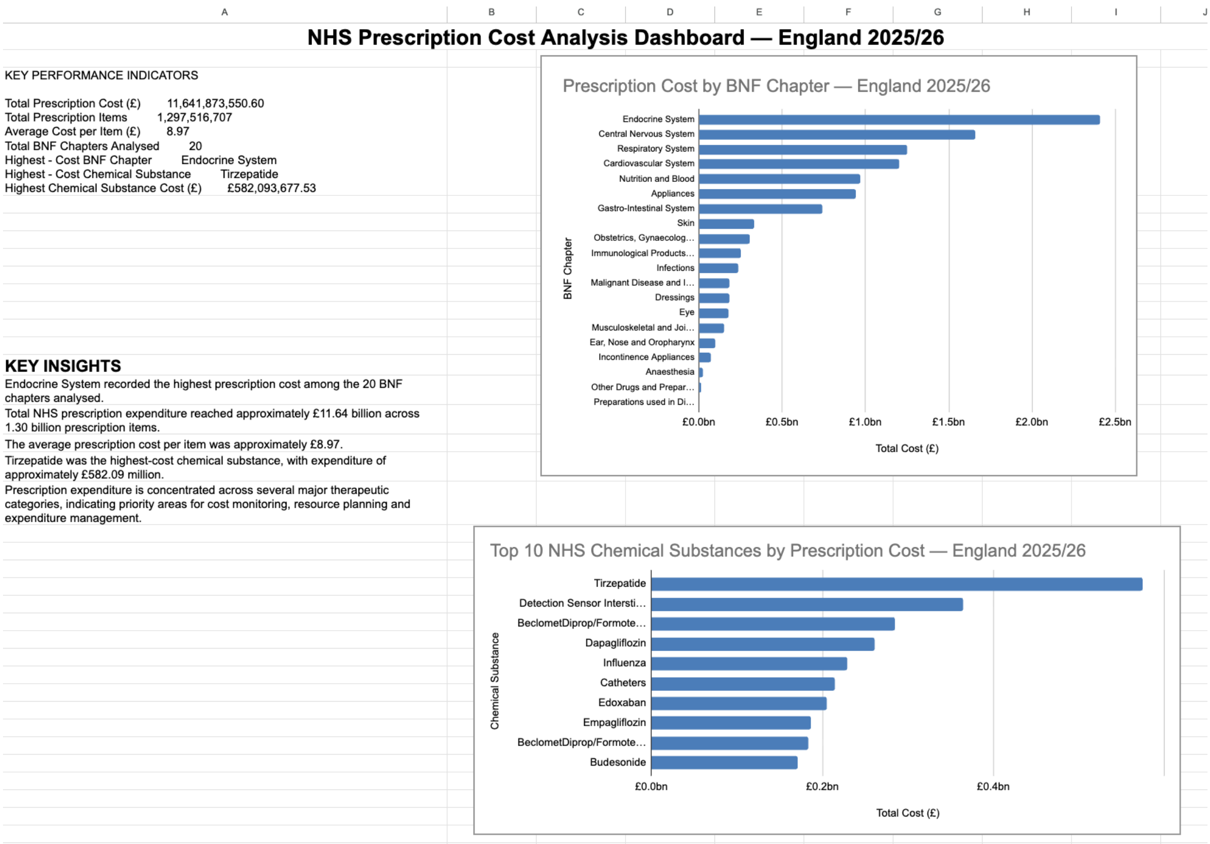
Dashboard Components

The dashboard contains:

- Total Prescription Cost
- Total Prescription Items
- Average Cost per Item

- Total BNF Chapters Analysed
- Highest-Cost BNF Chapter
- Highest-Cost Chemical Substance
- Highest Chemical Substance Cost
- Prescription Cost by BNF Chapter
- Top 10 NHS Chemical Substances by Prescription Cost
- Key analytical insights

The dashboard was designed to provide both an executive overview and sufficient category-level information to identify where further investigation may be required.



<u>Figure 1. NHS Prescription Cost Analysis Dashboard — England 2025/26</u>

9. Key Findings and Business Interpretation

9.1 Prescription expenditure is concentrated across major therapeutic categories

The **Endocrine System recorded the highest prescription expenditure among the 20 BNF chapters analysed**, making it the largest therapeutic expenditure category within the dataset.

The importance of this result lies not only in its ranking, but in the concentration of expenditure within particular therapeutic areas. When a category accounts for a substantial level of overall expenditure, changes in its prescribing volume, treatment mix or product costs can have a disproportionate effect on total prescribing expenditure.

The Endocrine System should therefore be treated as a **priority area for expenditure monitoring and further investigation**, rather than automatically as an area requiring cost reduction. High expenditure may reflect legitimate clinical demand, the volume of long-term treatment, the use of higher-cost products, or a combination of these factors. Further analysis would be required to separate these drivers and determine which are responsible for the observed expenditure.

9.2 £11.64 billion in expenditure reflects both the scale of prescribing and the financial impact of small changes

The analysed data represents approximately **£11.64 billion in prescription expenditure across 1.30 billion prescription items** during 2025/26.

The combination of high expenditure and very large prescription volume is particularly important. At this scale, even relatively small changes in average cost per item can translate into substantial changes in total expenditure. Similarly, an increase in prescribing volume within a widely used therapeutic category could materially increase overall costs even if individual items remain relatively inexpensive.

This means expenditure should be monitored through its underlying components rather than simply tracking the total amount spent. Separating changes caused by **volume, cost per item and product mix** would provide a clearer explanation of why expenditure is increasing or decreasing and would allow unusual movements to be investigated earlier.

9.3 The £8.97 average cost per item is a baseline for comparison, not an indicator of efficiency

Average prescription expenditure across the dataset was approximately **£8.97 per item**.

This figure provides a useful system-level reference point, but it should not be interpreted as a target or used independently to determine whether a therapeutic category is performing efficiently. Different BNF chapters contain treatments with fundamentally different cost structures, prescribing frequencies and clinical purposes.

Its greater value is as a **baseline for identifying variation**. Comparing cost per item across BNF chapters can highlight categories where expenditure is being driven more strongly by unit cost than by prescription volume. Over time, the same measure could also identify categories experiencing unusually rapid cost increases.

Combining cost per item with **total expenditure, prescription volume and expenditure share** would therefore provide a much more meaningful view of cost behaviour than comparing categories against the £8.97 overall average alone.

9.4 Prescription volume and expenditure reveal different types of cost pressure

The analysis shows that **high prescription volume does not necessarily correspond to the highest expenditure**, and expenditure rankings should therefore not be interpreted as prescribing-volume rankings.

A therapeutic category can generate high overall expenditure because a very large number of relatively inexpensive items are prescribed. Another category may have a lower prescription volume but still generate substantial expenditure because its individual products have a much higher average cost.

This distinction is important when investigating expenditure. A **volume-driven increase** may require investigation into prescribing demand and utilisation patterns, whereas a **cost-driven increase** may require examination of product prices, treatment mix or the growing use of higher-cost therapies.

For this reason, **total cost, prescription volume and cost per item should be assessed together**. The combination helps establish not simply where expenditure is high, but what is most likely to be driving it.

9.5 Tirzepatide is a significant substance-level cost concentration

Tirzepatide was the highest-cost chemical substance identified, with expenditure of approximately £582.09 million.

This result is important because it identifies a substantial expenditure concentration at a more granular level than the BNF chapter analysis. While chapter-level analysis identifies broad therapeutic areas requiring attention, chemical-substance analysis begins to identify the individual components contributing to those costs.

The £364.25 million expenditure should therefore be treated as a **trigger for deeper investigation rather than evidence of excessive spending**. The next stage would be to examine the number of items prescribed, average cost per item, changes in expenditure over time and the extent to which utilisation has changed.

Relevant clinical and operational context would also be required before drawing conclusions. A high-cost substance may represent increasing patient demand, adoption of newer technologies or treatments, higher product costs, or clinically appropriate changes in prescribing practice.

The finding therefore identifies **where further analysis should be concentrated without assuming that the expenditure itself is avoidable**.

9.6 Expenditure share provides a stronger basis for prioritisation than ranking alone

Ranking BNF chapters by total expenditure identifies which categories cost the most, but ranking alone does not show **how financially significant each category is relative to the overall £11.64 billion expenditure base**.

Calculating each chapter's percentage share of total expenditure adds this missing context. For example, two categories may appear close together in a ranking while representing materially different proportions of total expenditure. Expenditure share therefore provides a clearer measure of financial materiality.

This measure can also support a more structured monitoring approach. Categories representing a substantial share of expenditure could receive more frequent review, while significant increases in expenditure share could be flagged for investigation even before they materially alter the overall ranking.

Used alongside **total expenditure, prescription volume and cost per item**, expenditure share creates a more effective framework for identifying **where costs are concentrated, what is driving them and which areas should be investigated first**.

10. Business Recommendations

10.1 Introduce a prioritised expenditure-monitoring framework

Rather than applying the same level of monitoring to every BNF category, reporting should prioritize therapeutic areas representing the greatest expenditure and expenditure share.

A practical monitoring framework could classify categories according to:

Total Cost | Expenditure Share | Prescription Volume | Cost per Item

This would allow management attention to be directed toward categories with the greatest financial materiality.

10.2 Monitor expenditure drivers rather than expenditure alone

Changes in total cost should be decomposed into their underlying drivers.

Where expenditure increases, subsequent analysis should determine whether the movement is primarily associated with:

- Increased prescription volume
- Increased cost per item
- Changes in product or treatment mix

- High-cost individual substances
- A combination of these factors

This would improve the diagnostic value of management reporting and support more targeted investigation.

10.3 Establish exception-based KPI reporting

A recurring dashboard could monitor:

Total Prescription Cost | Total Items | Cost per Item | Expenditure Share

Rather than reviewing every category manually, thresholds could be introduced to flag significant changes in these indicators.

This would enable analysts and decision-makers to focus attention on material deviations and emerging expenditure pressures.

10.4 Conduct deeper analysis of high-cost chemical substances

High-cost substances such as Tirzepatide should be assessed using additional dimensions before management conclusions are made.

Further analysis should incorporate historical expenditure, prescription volumes, cost per item and relevant clinical or operational context.

The purpose would be to understand why expenditure is high rather than treating high expenditure itself as evidence of poor performance.

10.5 Extend the analysis into trend and regional reporting

The current project provides a strong expenditure baseline for 2025/26.

Future analysis should incorporate multiple financial years to identify:

- Persistent expenditure growth
- Changes in prescribing volume
- Changes in cost per item
- Emerging high-cost categories
- Shifts in expenditure concentration

Regional or Integrated Care Board-level analysis could subsequently identify geographical variation and provide more actionable operational insight.

11. Limitations

The project has several limitations that should be considered when interpreting the findings.

Single-period analysis: The project primarily evaluates the 2025/26 financial year and therefore does not establish whether observed expenditure patterns represent long-term trends.

Descriptive analysis: The analysis identifies patterns and cost concentrations but does not establish the clinical or operational causes behind them.

Clinical context: High expenditure should not automatically be interpreted as unnecessary or inefficient expenditure. Costs may reflect clinical need, treatment complexity, medicine prices, patient population and prescribing guidelines.

Aggregation: BNF chapter-level analysis provides a useful strategic overview but can conceal variation between individual medicines, products and prescribing organisations.

These limitations mean that the findings are most appropriately used to identify priorities for further investigation rather than to make direct clinical or cost-reduction decisions.

12. Skills Demonstrated

Business Analysis
Translated a broad healthcare expenditure problem into defined analytical questions, KPIs, findings and business recommendations.

Data Analysis
Analysed prescription expenditure, volume, cost per item and expenditure concentration to identify meaningful patterns and cost drivers.

Google Sheets
Used for data preparation, validation, calculations, exploratory analysis, KPI development, chart creation and dashboard reporting.

SQL
Developed queries involving aggregation, sorting, ranking, filtering and window functions to analyse prescription expenditure.

Google BigQuery
Structured and analysed healthcare data within a cloud-based analytical environment.

Data Visualisation
Developed a stakeholder-facing dashboard combining KPIs, category-level expenditure analysis and chemical-substance analysis.

Business Intelligence
Converted detailed prescription data into concise management information suitable for expenditure monitoring and decision support.

Healthcare Analytics
Applied analytical methods to NHS prescribing data while recognising the distinction between financial patterns and clinical decision-making.

Stakeholder Communication
Translated technical analytical outputs into concise findings and recommendations that can be understood by non-technical stakeholders.

13. Conclusion

This project demonstrates how NHS prescription data can be transformed into structured business intelligence by combining spreadsheet analysis, SQL and dashboard reporting.

The analysis identified approximately **£11.64 billion in expenditure across 1.30 billion prescription items**, highlighted significant variation across therapeutic categories, and identified major chapter- and substance-level expenditure concentrations.

Rather than treating high expenditure as an isolated performance issue, the analysis demonstrates the importance of evaluating **cost, volume, cost per item and expenditure share together** to understand the underlying drivers of prescribing expenditure.

The project demonstrates an end-to-end analytical workflow covering data preparation, KPI development, SQL analysis, data visualisation, business interpretation and evidence-based recommendations.

14. Project Resources

Project: NHS Prescription Cost Analysis — England 2025/26

Official Data Source: NHSBSA — Prescription Cost Analysis, England 2025/26:

https://www.nhsbsa.nhs.uk/statistical-collections/prescription-cost-analysis-england

Prescription Data Information:

https://www.nhsbsa.nhs.uk/prescription-data/dispensing-data/prescription-cost-analysis-pca-data

Data Provider: NHS Business Services Authority (NHSBSA)

Dataset: Prescription Cost Analysis — England 2025/26

Spreadsheet Analysis & Dashboard: Google Sheets

Database & SQL Analysis: Google BigQuery

GitHub Repository:

Portfolio Documentation: NHS Prescription Cost Analysis — England 2025/26

Tools & Techniques

Google Sheets | SQL | Google BigQuery | Data Cleaning | Data Analysis | KPI Development | Data Visualisation | Business Analysis | Healthcare Analytics | Business Intelligence